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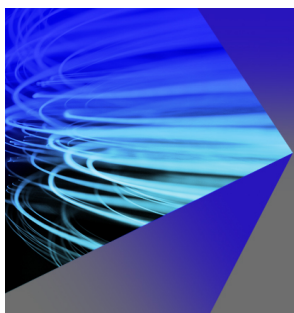
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Single-electron transistors in heterostructure nanowires

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Semiconductor-based single-electron transistors have been fabricated using heterostructure nanowire growth, by introducing a double barrier of InP into InAs nanowires. From electrical measurements, we observe a charging energy of 4 meV for the approximately 55 nm diameter and 100 nm long InAs islands between the InP barriers. The Coulomb blockade can be periodically lifted as a function of gate voltage for all devices, which is a typical signature of single-island transistors. Homogeneous InAs nanowires show no such effect for the corresponding voltage ranges. © 2003 American Institute of Physics. [DOI: 10.1063/1.1606889]

A single-electron transistor¹ (SET) is a device in which electrons tunnel one at a time through a small island connected to two leads via tunnel junctions. A potential applied to a nearby gate electrode controls the net charge on the island, allowing the device to be periodically changed between a conducting and a current-blocking state. Such transistors operate at temperatures where the thermal energy of the electrons, $k_B T$, is much smaller than the charging energy, e^2/C , of the device. For a typical transistor, fabricated with conventional lithographic techniques, this temperature is below that of liquid helium (4.2 K). Applications where the SET is of interest are, for example, sensitive charge detectors, electronics where low power dissipation is desired, and in single-electron turnstile devices. Today many different techniques are used to fabricate SETs in both metallic and semiconducting as well as in molecular systems.^{2–6}

Recently, considerable effort has been made to incorporate SETs into one-dimensional (1D) devices. One type of 1D device where SET effects have been studied extensively is carbon nanotubes, where poor contacts to a nanotube will cause the section of the nanotube between the electrodes to act as an island.⁷ The same effect has also recently been shown for InP nanowires, where high-resistive contacts result in an island coupled to the electrodes via tunnel barriers.⁸ In the nanotube system one can create a very small dot by inducing defects in a tube with, for example, scanning probe techniques,⁹ or by breaking a tube and contacting a nanoparticle with the two resulting tube leads.¹⁰ The former device can display single-electron charging effects even up to room temperature owing to the small size of the dot. However, the fabrication of such devices is generally too complex to allow use on a larger scale. One technique that has shown impressive results, and at the same time offers reproducibility, is the fabrication of vertical quasi-1D transistors by etching pillars in semiconductor substrates containing a double-barrier heterostructure.¹¹ This process, however, relies on a top-down approach, making it difficult to further scale down the devices.

In this letter we report on the use of a bottom-up approach for the fabrication of designed semiconductor SETs by epitaxial growth of InAs/InP heterostructured nanowires seeded from Au metal catalyst particles. The devices show ideal SET characteristics at low temperatures, where the Coulomb blockade of the current on the InAs island between the InP barriers can be periodically lifted as a function of backgate voltage. A comparison is made with electrical measurements on corresponding reference wires of homogeneous InAs where no charging effects occur. An important implication of our technique is that the transistor action is independent of contact design and therefore results in predictable and reproducible device characteristics.

N-type InAs nanowires with an InP double-barrier structure were grown by chemical beam epitaxy from size-selected gold catalyst particles deposited on a GaAs sample. Details of the growth can be found in Refs. 12–13. The wires had diameters between 55 and 65 nm and lengths of 3–4 μm , with roughly 5-nm-thick InP barriers and an InAs dot 80–100 nm long. The nanowires were transferred to a silicon sample with a 100-nm-thick SiO_2 layer, and metal electrodes were fabricated by electron beam lithography. The contact areas were etched in diluted $(\text{NH}_4)_2\text{S}_x$ for a few minutes prior to the evaporation of 20 nm Ni and 70 nm Au for the contacts. No postannealing was employed.

Homogeneous InAs wires, grown under the same conditions, with diameters greater than 50 nm, show linear I – V characteristics with resistances in the low- $k\Omega$ range from room temperature down to 4.2 K. We have observed that when the diameter of an InAs wire is less than 50 nm the resistance increases rapidly with decreasing wire diameter and temperature. We believe this effect is due to the large energy quantization in InAs as a result of the low effective electron mass, pushing the ground state of the lateral confinement to energies above the Fermi level. Figure 1 shows the conductance of homogeneous InAs nanowires of different diameters (105, 80, and 60 nm) as a function of backgate voltage, V_G .

The InAs/InP double-barrier devices studied here have a resistance in the $M\Omega$ range at room temperature as well as at 4.2 K. For a given wire diameter the resistance is very reproducible. Thinner wires from the same growth have a

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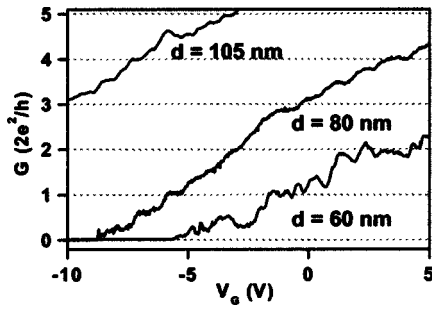


FIG. 1. Conductance plotted as a function of backgate voltage for homogeneous InAs nanowires with diameters of 105, 80, and 60 nm at $T=4.2$ K, showing the depletion of conduction electrons for negative gate voltages. The 105 and 80 nm plots were recorded using a two-terminal lock-in technique with an ac bias of 0.1 mV, and the 60 nm plot with a dc bias of 0.36 mV.

higher resistance, partly because of the smaller cross section, but more importantly due to thicker InP barriers as a result of a higher growth rate under the growth conditions used here. Similarly, thicker wires show a lower resistance. Selective InAs etching of wires in $(\text{NH}_4)_2\text{S}_x$ can be used to reveal the double-barrier structure, which is a helpful tool for transmission electron microscopy investigations in order to measure the island dimensions. The lower inset in Fig. 2(a) shows a wire after etching.

Electrical characterization (dc) at 4.2 K of the 55 nm diameter nanowire in Fig. 2(a) shows a Coulomb blockade in the $I-V$ characteristics with a charging energy of 4 meV, that can be lifted as a function of V_G [Fig. 2(b)]. The device shows ideal conductance oscillations, periodic in V_G , which is typical for a single-island system. Starting from the left in Fig. 2(c) each peak corresponds to the addition of one extra electron onto the island. The conductance oscillations are

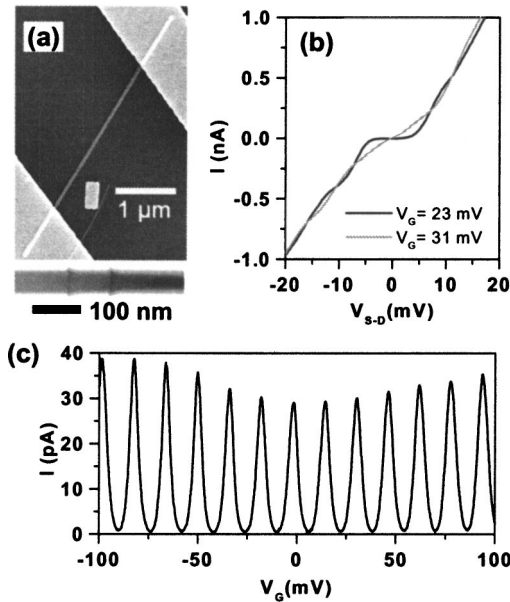


FIG. 2. (a) Scanning electron micrograph of an electrically contacted InAs nanowire with an InP double-barrier structure. The lower inset shows how a selective InAs etch can be used to reveal the InP double barrier (transmission electron micrograph). (b) $I-V$ curves for the 55 nm diameter device in Fig. 2(a) recorded for two different gate voltages at $T=4.2$ K, corresponding to a blocked (black) and a conducting state (gray). (c) Current through the device plotted as a function of the universal backgate voltage for $V_{SD}=0.5$ mV.

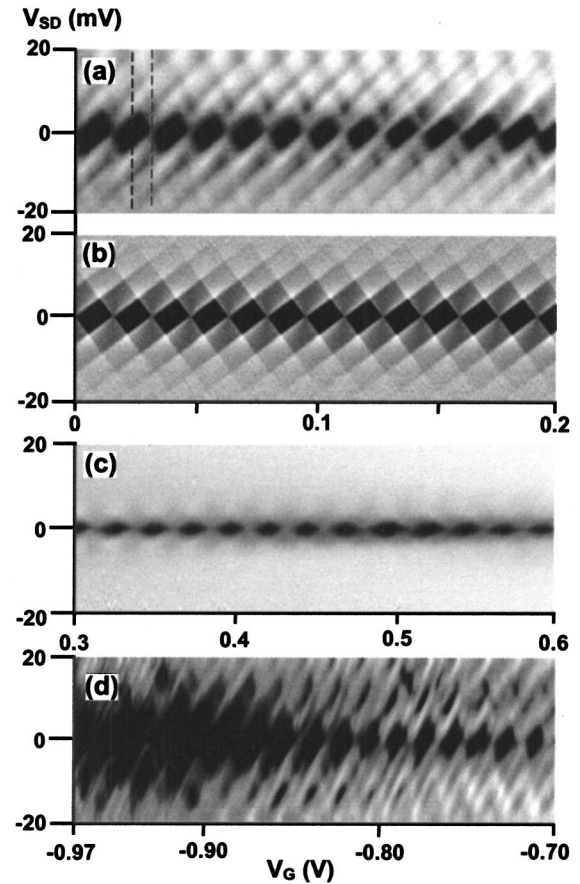


FIG. 3. dI/dV_{SD} plotted as a function of V_G and V_{SD} . Bright areas correspond to a higher value of dI/dV_{SD} . (a) Measurement at $T=4.2$ K for a 55 nm diameter device. The two vertical lines indicate the origin of the $I-V$ curves shown in Fig. 2(b). (b) Theoretical fit at $T=0$ K. (c) Measurement at $T=4.2$ K for a 65 nm diameter device for which the charging energy is reduced by a factor of 2. (d) Depletion of the nanowire in Fig. 3(a) for negative gate voltages, transforming the designed double-barrier device into a multi-island system.

visible up to roughly 12 K, for which $kBT=1$ meV. Figure 3 shows plots where the differential conductance, dI/dV_{SD} , is displayed in grayscale as a function of source-drain voltage, V_{SD} , and backgate voltage, V_G . Brighter regions correspond to a higher value of dI/dV_{SD} in these plots. A dI/dV_{SD} measurement of the wire in Fig. 2(a) is shown in Fig. 3(a). Based on a charging energy, $E_C=e^2/C$, of 4 meV, we estimate the total capacitance, C , to be 40 aF, and using $\Delta V_G=e/C_G$, the gate capacitance, C_G , to be 10 aF, where $\Delta V_G=16$ mV is the period for a Coulomb blockade region along the x axis. Figure 3(b) shows a theoretical fit at $T=0$ K of the electrical measurement in Fig. 3(a) where SIMON,¹⁴ a Monte Carlo based simulation program for nanostructures, has been used. A detailed analysis results in $C=36$ aF, and $C_G=10$ aF for this particular device. The capacitances in the tunnel junctions can be compared with that obtained for a corresponding plate capacitor, $C_P=\epsilon_r\epsilon_0 A/t$, where A is the area, $t=5$ nm, the thickness of the tunnel barrier, and $\epsilon_r=11.8$, the dielectric constant for InP. According to this simple formula a 55 nm diameter wire has a value of $C_P=50$ aF per junction, which gives a total capacitance of ~ 100 aF for the device, i.e., a factor of 2–3 larger than the value estimated from the charging energy.

A dI/dV_{SD} plot of a thicker nanowire, with a diameter of

65 nm, is shown in Fig. 3(c). Since A is larger and t is smaller for a thicker wire the capacitances of the tunnel junctions increase, resulting in a smaller charging energy (2 meV) for the island in this case. At the same time the current level increases by a factor of 150 mainly due to the reduction in t . Fitting with SIMON gives $C = 60$ aF and $C_G = 7$ aF. The reduced gate coupling is most likely due to the shorter length of the island.

For negative gate voltages, corresponding to the depletion of a homogeneous InAs nanowire as shown in Fig. 1, the Coulomb blockade structure becomes irregular due to the formation of a multi-island structure along the length of the wire, Fig. 3(d). The Coulomb blockade also becomes larger as a result of the addition of charging energies from the islands in series. However, for the normal operational range of the device, the conductance of the InAs nanowire itself is very insensitive to changes in the gate voltage, and the tunnel junctions will dominate the device resistance. Counting the number of Coulomb blockade diamonds from $V_G = 0$ V until the wire is depleted at $V_G = -0.8$ V, gives an estimate of the number of electrons confined on the island. Fifty diamonds, corresponding to the same number of electrons in a volume of $55 \times 55 \times 100$ nm³, indicates a conduction electron concentration of $\sim 10^{17}$ cm⁻³. However, here we have assumed that the island and the wire are depleted at the same time. Improved design for the gate would be obtained by having a capacitive coupling to the island only, as it would then be possible to deplete the island in a selective fashion.

In these devices we do not expect to see a negative differential resistance since the energy level splitting in the island is very small. This is in contrast to the resonant tunneling diodes reported in Ref. 13, with a dot length of only 15 nm, where the energy level splitting due to the quantum confinement is much larger than the charging energy. For that type of device a negative differential resistance is observed when the lowest mode in the nanowire emitter is pushed above the energy level in the dot.

In this letter, we have shown that SETs can be fabricated in a bottom-up approach using heterostructure semiconduc-

tor nanowire growth. For this type of device the active component is defined by the heterostructure itself, which means that the functionality is independent of contact design. However, a prerequisite for ideal device performance is low-resistive contacts to the nanowires in order to avoid charging effects and voltage drops in areas outside of the designed heterostructure. We have developed a contact technique for InAs nanowires which fulfills these requirements. The straightforward fabrication technique of these devices in combination with ideal electrical characteristics will probably allow more complex designs to be realized.

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